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CSC 428

Computer Modelling and Simulation (I)

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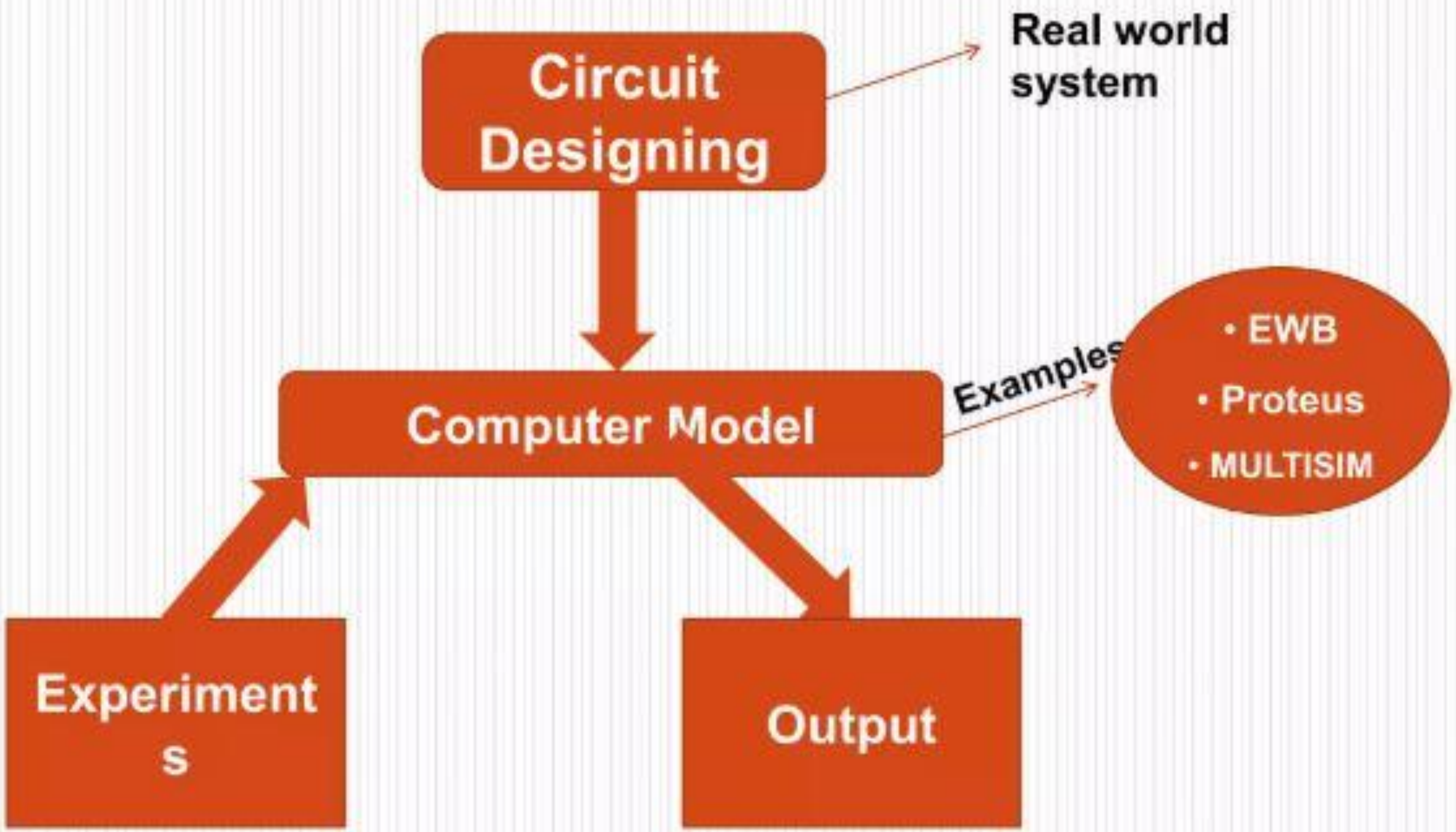
Contents

What is Computer Simulation?

Computer simulation is :

- The process to create an imitation of real world system or physical object into a computer model.
- Performing experiments to understand the behavior of system and evaluating new strategies.
- Then observing events, processes, properties and behavior of system, with computer model.

Example


Circuit Designing
Real world system
Computer Model
Examples
• EWB
• Proteus
• MULTISIM
Experiments
Output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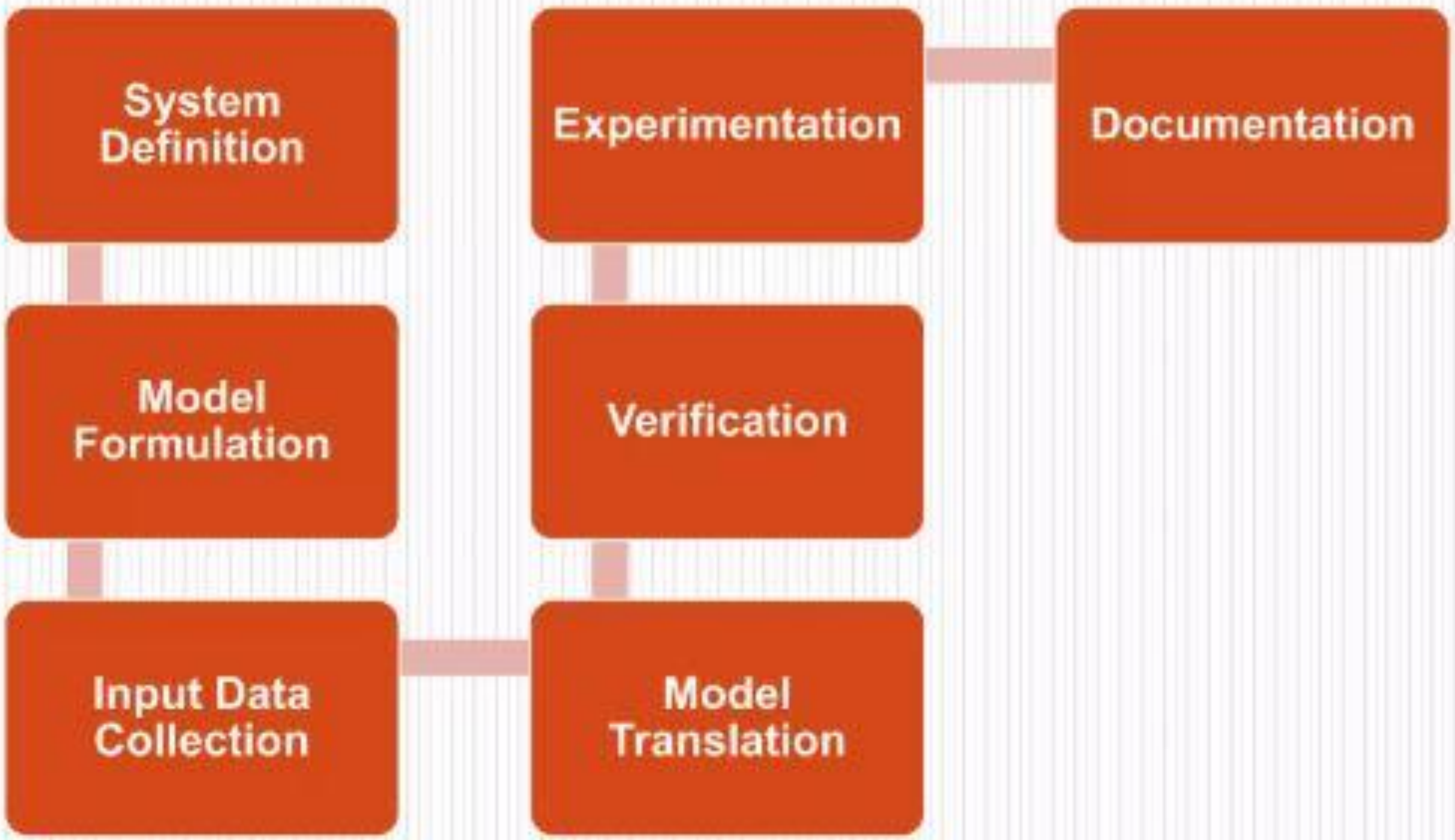
Process of Simulating a System



Applications of Computer Simulation

Weather Forecasting

- Computer simulation of weather forecasting can be used to predict storms

- The wind, rain pattern and temperatures e.t.c for the whole planet are simulated using very powerful computers.

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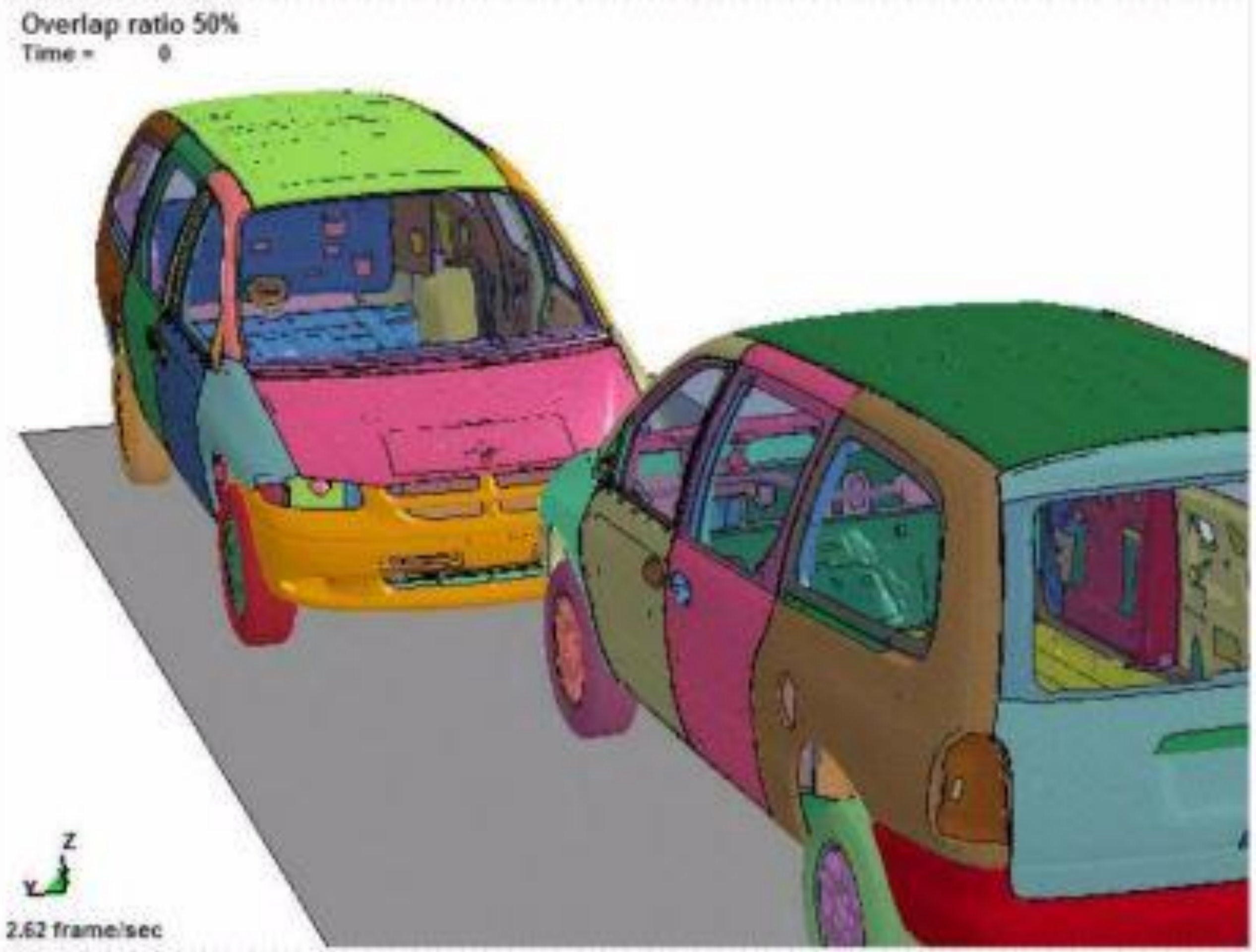
Simulation of car crash to study safer design

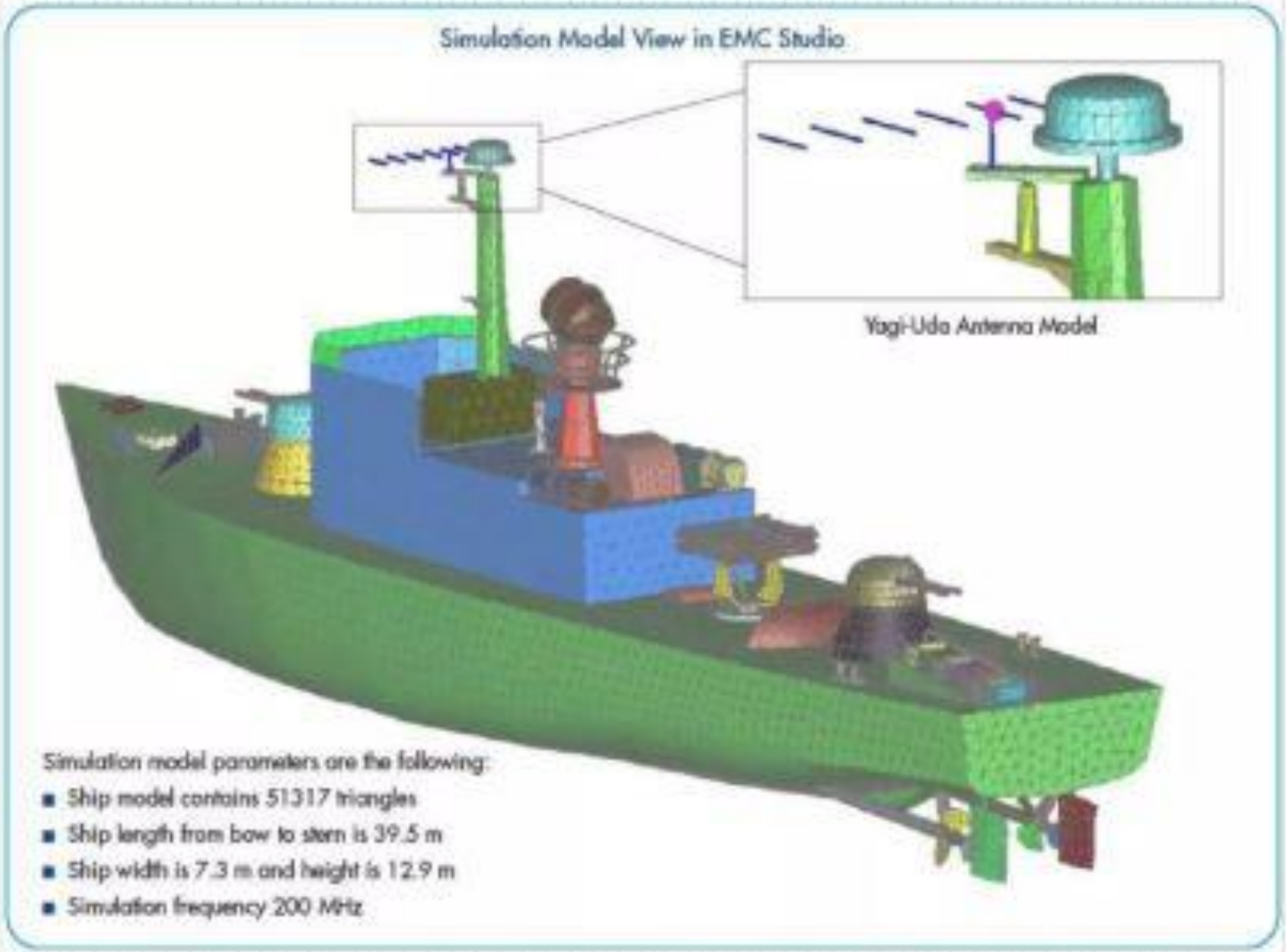


Designing

- Simulations software are used to design safer cars and improve their designs (performing real car crash just to check design stability is very expensive)

- Design of ships are also simulated first to achieve more perfection

Simulation of ship design

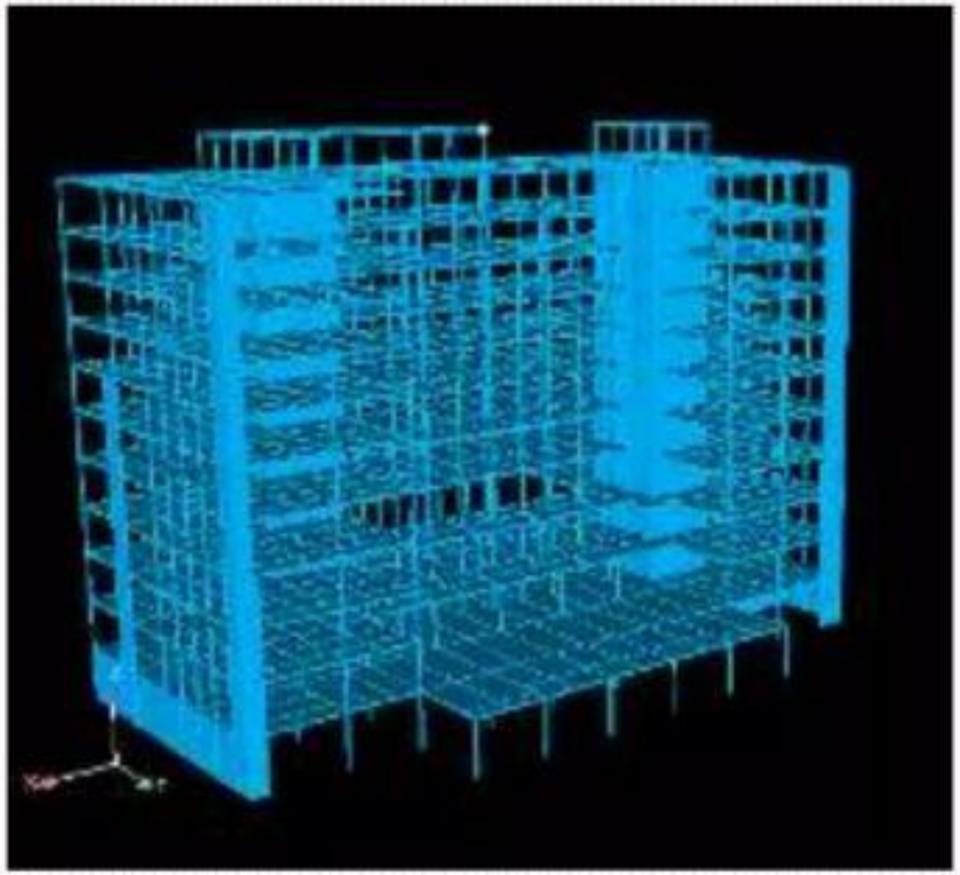


Architecture

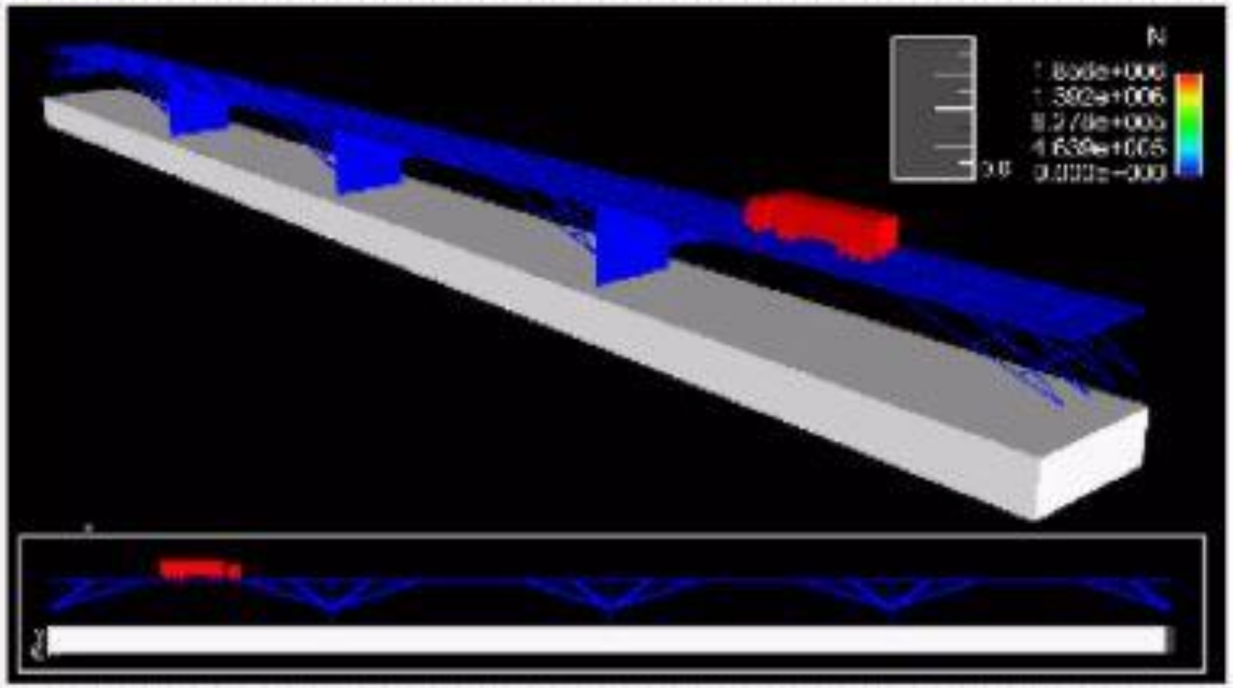
- Design stability of buildings are checked by simulation of earth quake through simulation software

- Bridges are also simulated to check how much load they can resist, and what improvement can be made

Earth quake simulation to check building stability



Bridge simulation



Aeronautics

- Trainee pilots have many hours of lesson in flight simulators before being allowed to fly an airplanes.
- These simulators provide realistic flying situation like storms, engine failure…
- This experience help them to tackle such situation in real, as they have gained a bit experience.

Flight simulation software

How many airplanes are flying in the world right now?

Live Web-based Simulation of Air Traffic

Visit www.flightradar24.com

Advantages of Computer Simulation

- To test a system without having to create the system for real (Building real-life systems can be expensive, and take a long time)

- To predict what might happen to a system in the future (An accurate model allows us to go forward in virtual time to see what the system will be doing in the future)

- To train people to use a system without putting them at risk (Learning to fly an airplane is very difficult and mistake will be made. In a real plane mistakes could be fatal!)

- To investigate a system in great detail (A model of a system can be zoomed in/out or rotated. Time can be stopped, rewound, etc.)

Disadvantages of Computer Simulation

- It depends upon the skills of the model maker
- Time consuming and high cost
- Sometimes, it does not give the best solution for system

Conclusion

- Computer Simulation and Modeling software are very useful in many areas.

- It gives great understanding of a system and also predicts the behavior of system in different situation